

ADITHYA RAJESH

AI / ML Developer · B.Tech CSE '29

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SUMMARY

B.Tech Computer Science student (Class of 2029) building intelligent systems across machine learning, computer vision, robotics and the web. Shipped a retrieval-augmented chatbot with citations, an NLP career-matching platform during an IBM internship, and gesture-driven interfaces. Comfortable end-to-end: data ? model ? API ? interface.

PROJECTS

RAG Campus Assistant

Python · FastAPI · ChromaDB · sentence-transformers · SSE

- ? Retrieval-augmented chatbot that answers campus questions strictly from institutional PDFs, with numbered inline citations.
- ? Built the full pipeline: PDF extraction ? overlap chunking ? MiniLM embeddings ? ChromaDB cosine search ? greedy-MMR diversification.
- ? Multi-provider LLM layer (Groq/OpenAI/Ollama) with token-level Server-Sent-Events streaming; retrieval-only fallback mode.
- ? Ships with a 20-test pytest suite covering chunking, metadata coercion and the streaming API contract.

NexaPath ? AI Career Copilot

Python · NLP · TF-IDF · Flask

- ? Career-copilot platform for Tier-2/3 India, built during an IBM SkillsBuild internship.
- ? TF-IDF + cosine-similarity matching engine aligning candidate profiles with opportunities; aligned to SDG 4/8/10.

Synth-Vision ? Gesture Interfaces

Python · OpenCV · MediaPipe

- ? Real-time hand-tracking interface manipulating physics-based UI elements purely through gestures.
- ? Currently expanding into a MediaPipe + pynput desktop control app (volume/scroll/slides).

DSA Python

Python · Jupyter

- ? Curated data-structures & algorithms collection with Jupyter-based analysis ? the foundation layer for all ML work.

SKILLS

LANGUAGES Python, C/C++, JavaScript

ML / DATA Pandas, NumPy, scikit-learn, model training & evaluation

COMPUTER VISION OpenCV, MediaPipe, real-time tracking

BACKEND / WEB FastAPI, Flask, Node.js, REST APIs, SSE streaming

HARDWARE / ROBOTICS Arduino, sensors, actuator logic

TOOLS Git/GitHub, Jupyter, pytest, ChromaDB

EDUCATION

B.Tech, Computer Science and Engineering

TKM College of Engineering, Kollam 2025 ? 2029 (expected) · APJ Abdul Kalam Technological University

ADDITIONAL

- ? IBM SkillsBuild internship ? AI career platform track
- ? Build-in-public roadmap: sign-language translator and a LoRA fine-tuned Manglish LLM